Gains on Synthetic Occlusion Are Not Enough: A Controlled Study of the Transfer to Real Occlusion in Person Detection

Celia Patricia Apaza Hilasaca, Brayan Raul Condori Quispe, Jhan Logan Ramos Quispe
Universidad Peruana Unión (UPeU), Juliaca, Peru

***Abstract*—Synthetic occlusion augmentation is used to make person detectors robust to partial visibility, yet its benefit is usually measured on the same artificial occlusion. We ask whether that gain is enough to expect one on real occlusion in other datasets. In 21 controlled blocks (133 runs), a baseline and an augmented model share data and hyperparameters; the blocks vary the occluder, the detector (six, one-stage, two-stage and transformer), the domain and the augmentation dose, and are measured on a synthetic control, on two sources of annotated real occlusion (by day and by night) and on an approximate proxy in ExDark, with hypotheses registered internally with a cryptographic fingerprint before training. In the 20 blocks with a control, augmentation improves that control (+2.6 to +60.9 AP@0.5 points). With the usual dose and equal exposure (12 blocks), a post hoc mixed model estimates between -1.1 and +1.0 points depending on the source, hardly compatible with a mean gain of 2 points. Only Faster R-CNN improves two real sources: at the usual dose it did not meet the rule again with five seeds, and at a high dose it improves unoccluded people equally. With 1,183 fine-tuning images from COCO, pixel-level augmentation and AP@0.5, passing a synthetic control is not enough to claim robustness to real occlusion.**

***Index Terms*—person detection, occlusion, data augmentation, robustness, low light, internal register with a cryptographic fingerprint**

I. Introduction

Person detection with RGB cameras is used in video surveillance and mobile robotics, which also operate at night. In that setting, occlusion, a major cause of pedestrian-detection failures [1], and low light [2] coincide. In our data, the baselines lose 0.39 to 0.62 AP@0.5 on occluded people relative to visible ones, by day and by night (Section IV).

A widespread remedy is *synthetic occlusion augmentation*: during training, parts of the person are erased or covered [3]–[5]. Its benefit is usually measured on the same artificial occlusion or, when real occlusion is used, within the training dataset (CityPersons, CrowdHuman [6]); rarely on genuinely occluded people in other datasets. If the synthetic gain does not carry over, synthetic benchmarks overstate the robustness of the surveillance systems validated on them. In Peru, municipal security services had 14,722 operational video-surveillance cameras in 2023 [7]; with no public Peruvian dataset of people occluded at night that we know of, those systems rely on models trained on foreign data, like those used in this study.

Gap. In the literature reviewed (Section II) we found no study that (i) measures the effect on annotated real occlusion with a synthetic control confirming that the augmentation was learned, (ii) with several real sources, (iii) tests alternative explanations (occluder, detector, domain, data volume, exposure, dose) with controlled experiments and (iv) fixes hypotheses and rules before seeing the results.

Question, objectives and contribution. Is a gain on synthetic occlusion enough to expect a real gain? The general objective is to determine whether synthetic occlusion augmentation that improves a synthetic control also improves the detection of genuinely occluded people. The specific objectives are to measure the baseline's degradation under occlusion (OE1, AP@0.5 by stratum); the effect of augmentation (V3−V1) by occluder and dose (OE2), detector (OE3), and training domain and data (OE4); and to explore its mechanism (OE5: specificity, visible box and occluder realism). We contribute (1) controlled evidence in 21 blocks with five kinds of occluder, six detectors and three doses; (2) a global mixed-model estimate and a specificity analysis showing that passing the synthetic control does not guarantee a real gain, and that the only transfer that passes the rule on two sources, at a high dose and with 3 seeds, is not specific to occlusion; and (3) a minimum protocol (positive control, V1 and V3 on the same machine, fixed epochs, 5 seeds, several sources, specificity and a prior rule), met in full only by H18a. Each 3-seed block costs 0.5–2.6 GPU hours on an RTX 4090.

II. Related work and explanatory framework

Detection under occlusion and low light. Pedestrian detectors tackle occlusion through architecture and evaluate within one benchmark: by parts or visible boxes [8]–[10] or by fusing levels with Random Erasing [11]; at night, a YOLOX with domain transfer improves without stratifying by occlusion [12].

Augmentation as a training signal and as a test. Random Erasing [3] inspired pedestrian variants evaluated within CityPersons or CrowdHuman [4], [5]; Copy-Paste [13] and A-Fast-RCNN [14] paste objects or occlude features without measuring carry-over to other datasets. Synthetic-corruption benchmarks [15] measure how much a detector withstands without checking whether gains carry over. Robustness to synthetic shifts barely carries over to natural shifts [16]; it does in some cases [17] and depends on the similarity between augmentation and test [18]. Synthetic occlusion *ranks* already-trained classifiers much as real occlusion does [19], and occlusion augmentation does not solve occlusion in pose

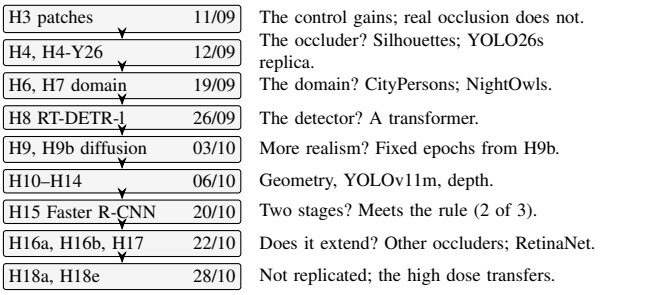


Fig. 1. Adaptive programme: each block answers an alternative explanation of the previous one (2026 dates on which the hypothesis was fixed).

estimation [20]. For pedestrians, style-transfer augmentation was evaluated across datasets without stratifying by occlusion [21], and a review places augmentation alongside part-based models and amodal supervision [22]. Compositional networks represent objects by parts [23], [24]; WALT learns amodal representations from real occlusion [25]. As a framework we use shortcut learning [26]: a network can reduce its loss by learning the artificial occluder without learning to detect incomplete people, and pedestrian detectors already generalize poorly across datasets [27], [28].

III. Method

Design, hypotheses and rule. The unit of analysis is the cell: one block (a V1–V3 pair) evaluated on one source. Each block is labelled with its registered hypothesis (H3–H18), so it can be traced to the register. In each block, V1 and V3 share data, hyperparameters, base augmentation and machine, except H16, whose V1 are those of H15, trained on another machine of the same model; V3 adds occlusion augmentation only in training. H3 and H4 predicted that the synthetic gain would carry over to real occlusion; the later transfer hypotheses predict that it does not appear when the occluder (H9, H9b, H10, H14; OE2), the detector (H8, H12, H15, H17; OE3), the domain and data (H6, H7; OE4) or the dose (H18; OE2) change. The prediction holds (transfer *not detected*) if fewer than two of the three real sources *pass*, i.e., improve AP@0.5 by at least +2 percentage points (pp, mean over seeds) with a 95 % image-level CI that excludes zero, and the synthetic control does pass; if the control fails, the block is inconclusive. H11 predicted the opposite (a gain on the most occluded CityPersons people), H16a and H16b that the Faster R-CNN transfer would extend to other occluders, and H13 a realism ordering of the occluders (OE5). Each hypothesis, its rule and its code were registered internally with a SHA-256 cryptographic fingerprint before training its block and none was reformulated; the fingerprint shows that the files did not change, but the project itself recorded their date (external timestamp and Zenodo deposit, DOI 10.5281/zenodo.24122872, only since 10 November 2026). The programme is adaptive (Fig. 1): the question was reoriented after the first experiment, H9b–H17 were designed after seeing earlier results and H18 answers an external review. Anything marked as post hoc decides no hypothesis.

Data. ExDark [2], with the split of Cui et al. [29], has 2,658 images with people; 1,183 are used for training. Real-occlusion sources: (1) NightOwls [30], at night and from a vehicle, with its authors' binary occlusion attribute (530 occluded pedestrians in 2,000 validation frames); (2) CityPersons [31], by day, *heavy* (visibility 20–65 %, height $\geq$50 px); (3) a proxy of occlusion between people in ExDark from box overlap. The proxy was adopted after H3 and H4 (13/09/2026), because the planned manual annotation was unreliable (κ 0.06–0.60 between passes), and its validity is low (Spearman and κ of 0.24 against CityPersons visibility); from H6 on it was fixed before training, and we repeat the analysis without it (Section IV). It is a secondary source; it stays in the "2 of 3" rule because it was registered that way from H6 on. CityPersons and NightOwls are scored with the full-body box, as in their protocols, and ExDark with the box of its annotations; a synthetically occluded person keeps its full box and pasted person silhouettes are labelled (up to 0.13, 0.5 and 1.75 people per image depending on the dose). All three datasets are used under non-commercial research licences.

Augmentation, dose and control. Occluder families: patches; silhouettes of real objects and people; diffusion (Stable Diffusion 1.5 [32]); silhouettes with the real visibility geometry of CityPersons; and depth (Depth Anything V2 [33]). The usual dose occludes an image with probability 0.5, and in it 1–3 people with coverage from 0.30 to 0.85 (0.40–0.95 with the real geometry; for diffusion and depth, the occluded image replaces the original with probability 0.5). H18e changes only the silhouette dose: low (probability 0.25, one person, coverage 0.15–0.40) and high (1.0; 1–6 people; 0.50–0.95). The *control* applies the same generator to every person in the test set of the training domain with coverage from 0.70 to 1.00 and unseen occluders; the cross-family matrix uses the other families' controls (parameters in the supplement).

Detectors and exposure. YOLOv11s, YOLOv11m [34], YOLO26s [35], RT-DETR-l [36] and, from torchvision (ResNet-50 FPN), Faster R-CNN [37] and RetinaNet [38], pretrained on COCO [39], at 640 px on the long side (recipes in the supplement). In H4, H4-Y26, H6 and H8 each variant was selected by validation, with *unequal* exposure (V1 at epochs 1–51, V3 at 31–89); from H9b on, V1 and V3 are trained for 30 fixed epochs (60 in H11) and evaluated with the last weights: these 14 blocks are the primary evidence. Each variant is trained with 3 seeds and H18a with 5 (H9b and H10-s were extended to 5 separately; here their seeds 0–2 are used). In total there were 133 runs and 109 GPU hours (log in the supplement). Machines: a laptop with an RTX 5050 (8 GB) and, for H11, H12 and H14–H18, a rented RTX 4090. V3 barely improves on validation between epochs 25 and 30 ($<$0.6 pp), and H11, with 60 epochs, does not transfer either (supplement).

Evaluation. AP@0.5 of *person* by occlusion stratum, with the other strata as *ignore* regions [31]; it is the only metric common to the three sources and the control. CIs by *bootstrap* over images (1,000 replicates) [40], which decides the rule, and hierarchical over seeds and images (2,000), which we

TABLE I

BLOCKS, HYPOTHESES AND RESULTS: CHANGE IN AP@0.5 (PP) OF V3 OVER V1, MEAN OF 3 SEEDS (5 IN H18A). EXP.: EQUAL (E), UNEQUAL (U) OR EARLY (EA) EXPOSURE. NO: NIGHTOWLS OCCLUDED ($n = 530$); CP: CITYPERSONS *heavy* ($n = 733$); EX: EXDARK PROXY, SECONDARY SOURCE ($n = 101$); CONTROL: 684–2,131 PEOPLE. IN BRACKETS, HIERARCHICAL 95 % CI (SEEDS AND IMAGES). * PASSES THE REGISTERED RULE (+2 PP, IMAGE CI > 0). ‡ RULES OUT +2 PP; WITHOUT ‡, "NOT DETECTED" DOES NOT PROVE ABSENCE (POWER $\approx 1/3$). VERDICTS: *Not detected (k of 3)*, FEWER THAN 2 SOURCES PASS; *Detected*, 2 OR 3 (PREDICTION REFUTED); *Not confirmed* OR *Refuted*, IT PREDICTED A GAIN OR TRANSFER AND THERE WAS NONE; *Inconclusive*, NO CONTROL; *Same pattern*, REPLICATION OF H4; H10-S, ARM OF H10. † DOES NOT MEET THE RULE AGAIN WITH 5 SEEDS (H18A).

Block	Detector	Training	Occluder	Exp.	Control	NO	CP	EX	Verdict
Exploratory: unequal (U) or early (Ea) exposure									
H3	YOLOv11s	ExDark	Patches	Ea	+60.92*	−3.17 [−7.9; +2.1]	−0.36‡ [−2.4; +1.6]	−3.25‡ [−7.1; +0.9]	Not confirmed (predicted a gain)
H4	YOLOv11s	ExDark	Silhouettes	U	+10.63*	−16.79‡ [−20.4; −13.0]	−3.94‡ [−7.1; −1.0]	−5.02‡ [−11.9; +1.3]	Not confirmed (predicted a gain)
H4-Y26	YOLO26s	ExDark	Silhouettes	U	+12.01*	−14.31‡ [−18.1; −10.1]	−4.42‡ [−6.9; −1.6]	−3.51‡ [−9.5; +1.9]	Same pattern
H6	YOLOv11s	CityPersons	Silhouettes	U	+23.74*	−1.35‡ [−3.5; +0.7]	+1.82 [−0.9; +4.4]	+0.94 [−4.1; +6.7]	Not detected (0 of 3)
H7	YOLOv11s	NightOwls 20,000	Silhouettes	U	—	−1.43‡ [−4.5; +1.6]	−3.94‡ [−8.2; +0.8]	−9.15‡ [−17.2; −3.3]	Inconclusive (no control)
H8	RT-DETR-l	ExDark	Silhouettes	U	+10.97*	−6.98‡ [−11.2; −2.8]	−3.47‡ [−6.3; −0.7]	−2.21 [−7.6; +3.4]	Not detected (0 of 3)
H9	YOLOv11s	ExDark	Diffusion	Ea	+2.60*	−1.93‡ [−3.9; −0.1]	+1.23 [−1.7; +4.0]	+2.82* [−1.2; +7.4]	Not detected (1 of 3)
Primary evidence: equal exposure (E)									
H9b	YOLOv11s	ExDark	Diffusion	E	+15.23*	+0.01‡ [−1.6; +1.8]	+1.52 [−0.0; +3.4]	−3.03 [−8.8; +2.9]	Not detected (0 of 3)
H10	YOLOv11s	ExDark	Real geometry	E	+18.39*	−2.05‡ [−4.6; +0.3]	+2.37* [+0.6; +4.6]	−4.88‡ [−11.6; +1.0]	Not detected (1 of 3)
H10-s	YOLOv11s	ExDark	Silhouettes	E	+10.99*	−1.12‡ [−2.6; +0.5]	+2.26* [+0.7; +4.0]	−3.50‡ [−9.6; +1.9]	(H10 arm)
H11	YOLOv11s	CityPersons	Silhouettes	E	+24.66*	+1.95 [−0.2; +4.7]	+1.31 [−0.8; +3.5]	−2.44 [−7.3; +3.2]	Not confirmed (predicted a gain)
H12a	YOLOv11m	ExDark	Silhouettes	E	+14.45*	−0.39 [−3.9; +2.8]	+0.25 [−1.3; +2.1]	+3.04 [−2.6; +8.3]	Not detected (0 of 3)
H12b	YOLOv11m	ExDark	Diffusion	E	+19.04*	+0.78 [−3.2; +4.3]	+1.11 [−0.5; +2.9]	−0.56 [−5.8; +5.1]	Not detected (0 of 3)
H14	YOLOv11s	ExDark	Depth	E	+7.42*	−1.50‡ [−3.4; +0.3]	−0.16‡ [−1.9; +1.3]	+5.49* [−0.3; +11.1]	Not detected (1 of 3)
H15	Faster R-CNN	ExDark	Silhouettes	E	+14.72*	+3.28* [+0.7; +5.8]	+2.33* [+0.1; +4.9]	−1.00 [−6.7; +4.0]	Detected (2 of 3)†
H16a	Faster R-CNN	ExDark	Diffusion	E	+20.35*	+0.92 [−1.6; +3.6]	−0.76‡ [−2.6; +1.2]	+0.80 [−5.1; +6.3]	Refuted (predicted transfer)
H16b	Faster R-CNN	ExDark	Depth	E	+9.63*	+0.37 [−2.1; +2.6]	+0.60 [−1.6; +2.8]	−0.45 [−6.7; +5.6]	Refuted (predicted transfer)
H17	RetinaNet	ExDark	Silhouettes	E	+6.60*	−2.11 [−7.2; +2.8]	−0.28‡ [−2.1; +1.5]	−3.77 [−9.9; +2.8]	Not detected (0 of 3)
Primary evidence: replication of H15 and dose (H18)									
H18a	Faster R-CNN	ExDark	Silhouettes	E	+14.44*	+4.26* [+2.2; +6.3]	+1.74 [−0.1; +3.7]	−2.20 [−6.9; +2.9]	Not detected (1 of 3)
H18e-l	Faster R-CNN	ExDark	Silhouettes, low dose	E	+2.94*	+2.90* [+0.8; +4.9]	+1.74 [−1.1; +4.1]	−1.91 [−6.6; +3.2]	Not detected (1 of 3)
H18e-h	Faster R-CNN	ExDark	Silhouettes, high dose	E	+25.62*	+8.59* [+5.7; +11.3]	+3.78* [+0.9; +6.3]	+0.60 [−5.2; +6.5]	Detected (2 of 3)

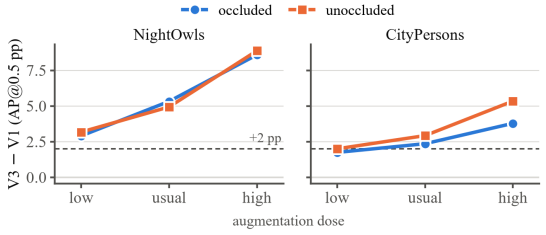


Fig. 2. Augmentation dose (H18, Faster R-CNN, silhouettes; seeds 0–2; Table I gives H18a with 5): V3−V1 for occluded people (CityPersons *heavy*) and unoccluded people (*bare*). Dashed: +2 pp.

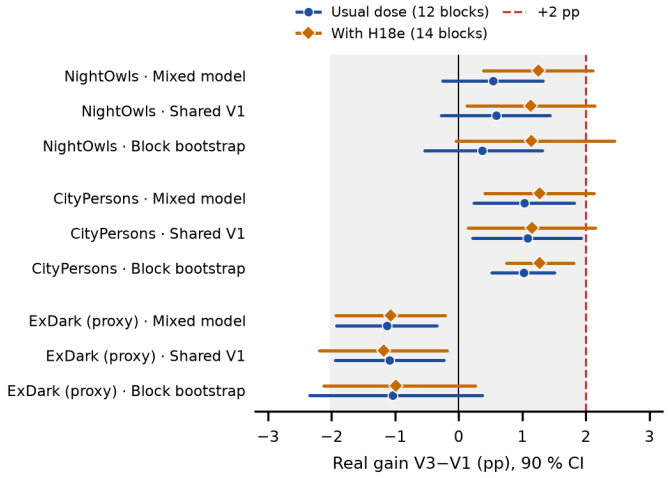


Fig. 3. Global V3−V1 estimate per source with 90 % CI (equivalence test, ±2 pp margin; post hoc analysis) under three specifications: the mixed model in the text, with a component for V1 shared across blocks, and a block bootstrap. Blue: usual dose (12 blocks); orange: with the H18e dose blocks (14). Grey band: ±2 pp.

apply to every cell as the uniform between-seed criterion. The +2 pp threshold is the smallest effect of interest: it equals, for Faster R-CNN, about 17 more occluded people detected per 1,000 at 0.1 false positives per image (post hoc analysis), and exceeds the between-seed SD of V1 (median 1.5 pp). Not detecting transfer does not prove its absence: with a true +2 pp on all three sources the rule detects it in one block out of three (with +3 pp, in 94 %). A cell *rules out* +2 pp if the upper limit of its CI is below +2 [41]. As a global estimate (post hoc analysis) we fit a mixed model by REML to the per-seed differences: a fixed effect per source and random effects for the block and for the seed within the block. Sensitivity: a component for V1 shared across blocks, a block bootstrap and TOST (90 % CI, ±2 pp margin). Secondary metrics (supplement) agree: with MR^{-2} [1], 4 of 51 real cells improve (3 on ExDark); with AP@0.5:0.95, none of 24 in the first eight blocks.

IV. RESULTS

Degradation (OE1). In the 10 baselines trained on ExDark with equal exposure, AP@0.5 goes from 0.61–0.69 unoccluded to 0.15–0.21 on occluded NightOwls people, from 0.47–0.65 to 0.08–0.15 on CityPersons *heavy* and from 0.78–0.85 to 0.20–0.31 on severe ExDark occlusion.

Global estimate (OE2–OE4). All figures are V3−V1 in AP@0.5 (Table I). All 20 controls pass, from +2.6 to +60.9 pp. With the usual dose and equal exposure (12 blocks, 114 per-

seed differences), the mixed model estimates +0.54 [−0.40; +1.48] on NightOwls, +1.03 [+0.09; +1.97] on CityPersons and −1.13 [−2.08; −0.19] on ExDark (95 % CI): with the 90 % CI (TOST; Fig. 3), the upper limits are +1.33, +1.82 and −0.34, hardly compatible with a mean gain of 2 pp; the CityPersons limit is +1.93 with shared V1 and +1.51 with a block bootstrap, so no specification reaches +2 at the usual dose. Adding the block's gain on its control does not change the estimate (+0.16 pp per SD [−0.55; +0.87]; per-source scatter in the supplement). Including the dose blocks (H18e), the 90 % upper limits rise to +2.11 on NightOwls and +2.13 on CityPersons, above the threshold. Descriptively, of those 36 cells 5 pass with the hierarchical CI, 9 rule out +2 pp and 22 are inconclusive; over all usual-dose blocks with a control, 5 of 54 pass, against 1.2 expected by chance: gains are few, but more than chance would give.

Exposure. When V1 and V3 are selected at different epochs, the mean effect is −4.94 pp; the harm of silhouettes on NightOwls (−16.8 in H4) shrinks to −1.12 with fixed epochs (H10-s): unequal exposure explains the harm, but equalizing it produces no gain.

Detector, domain and data (OE3, OE4). The pattern repeats with RT-DETR-l (H8), YOLO26s and YOLOv11m (H12). Faster R-CNN with silhouettes (H15) passes the rule on NightOwls (+3.28) and CityPersons (+2.33), also with the hierarchical CI ([+0.67; +5.81] and [+0.13; +4.92]). The 5-seed replication on another machine (H18a) keeps NightOwls (+4.26 [+2.21; +6.32]) but not CityPersons (+1.74 [−0.08; +3.69], compatible with the +2.33 of H15): 1 of 3. The tests that bound H15 do not transfer either: with diffusion and depth (H16a, H16b), Faster R-CNN learns the augmentation (+20.35 and +9.63) and no real cell exceeds +0.92, and RetinaNet with silhouettes (H17) gives 0 of 3, with a V3 that worsens on all three real sources. Training in the same domain is not enough either: within CityPersons (H6, H11) the control gains about +24 and *heavy* less than +2.

Dose (OE2). With Faster R-CNN, the gain grows with the dose on the control (+2.9, +14.8 and +25.6 pp), on NightOwls and on CityPersons, but not on ExDark (Fig. 2; ExDark in Table I). The high dose passes on two sources, also with the hierarchical CI (H18e-h), but it is not specific to occlusion: unoccluded people gain as much or more (+8.87 and +5.34), and overall ExDark AP drops by 1.98.

Occluder (OE2). Neither diffusion, nor real geometry, nor depth meets the rule; in the cross-family matrix, 13 of 20 cells pass.

Post hoc analyses (OE5). *Without ExDark*, requiring NightOwls and CityPersons, only Faster R-CNN with silhouettes (H15) and its high dose (H18e-h) detect transfer; at the usual dose, only H15, which does not meet the rule again with five seeds. *Specificity:* subtracting the gain on unoccluded people, of the 6 cells in H3–H17 that pass the rule (image CI), only the two from the ExDark proxy are specific, and none withstands the hierarchical CI. *Visible box:* scoring CityPersons with the visible box, the *heavy* gain exceeds +1.3 in no block of H3–H17 (H15: +0.74): part of what is gained is estimating the full extent of the person. *Realism* (H13, partially confirmed): with DINOv2 [42], each family's occlusion is told apart from real CityPersons occlusion (AUC 0.94–0.99) more easily than an unoccluded person (0.88).

V. Discussion

As in Taori et al. [16] and Pytel et al. [20], what is gained against a synthetic perturbation barely carries over to the real one; this does not contradict Kassaw et al. [19] (synthetic occlusion can rank models and still not suffice as a training signal) and, as in Hendrycks et al. [17] and Mintun et al. [18], carry-over depends on the augmentation–test pair and on the dose. The high dose improves the external sources for visible and occluded people alike; regularization against domain shift would explain it, as would the extra positives (up to ×3.5 pasted, labelled people, as in Copy-Paste [13]), and neither reading is robustness to occlusion. That passing the control is not enough is compatible with a shortcut [26]: the artificial occluder remains distinguishable from the real one (H13), although we did not test this confirmatorily. For the 14,722 municipal cameras [7], validating a model on synthetic occlusion alone does not guarantee that it detects occluded people at night.

Limitations. *Scope:* 1,183 training images, fine-tuning from COCO and three doses tested only with Faster R-CNN and silhouettes; at another scale or dose the result may change. *Data:* almost all real occlusion is in datasets other than the training one; CityPersons is daytime, NightOwls is recorded from a vehicle and the ExDark proxy is secondary (III); everything is single-class AP@0.5 at 640 px. *Protocol:* three seeds in most blocks; cells and blocks share baselines (sensitivity in IV), with no multiplicity correction (comparison with chance in IV); unequal exposure in the first blocks; the mixed model is post hoc, and the fingerprints only have an external timestamp (OpenTimestamps) since 10 November 2026. *Future work:* a dose curve with other detectors, at larger scale and without labelling the pasted people; occlusion mined from real images and amodal supervision [25]; night-time domain adaptation and part-based architectures [24].

VI. Conclusion

Real occlusion degrades the baseline in all three sources (OE1). Augmentation improved detection on synthetic occlusion in the 20 blocks with a control. On real occlusion, with the usual dose and equal exposure (12 blocks), the (post hoc) mixed model is hardly compatible with a mean gain of 2 pp on all three sources (OE2–OE4), also with shared V1 or a block bootstrap, though not when the doses are included; the only combination that passed the rule, Faster R-CNN with silhouettes, did not meet the rule again with five seeds. A high dose does improve NightOwls and CityPersons, but equally for unoccluded people, so it is not robustness specific to occlusion. The synthetic occluder remains distinguishable from the real one (OE5). With pixel-level augmentation, 1,183 training images and single-class AP@0.5, passing a synthetic control is not enough to claim robustness to real occlusion:

it takes annotated real occlusion, V1 and V3 on the same machine, five seeds and a specificity criterion.

Declarations

Data and code. Code, logs, fingerprints, per-cell data and supplement (with the literature search) at https://github.com/jhanrq8888/argos-occlusion-transfer (public); the weights of the primary evidence (67 runs, with their SHA-256 in the repository) are on Zenodo, DOI 10.5281/zenodo.24120199 (CC BY-NC 4.0). **Funding:** none. **Competing interests:** none. **Contributions (CRediT):** J. L. Ramos Quispe: conceptualization, methodology, software, formal analysis, investigation, data curation and writing of the original draft; C. P. Apaza Hilasaca: conceptualization, methodology (design of the pre-specification), review and editing; B. R. Condori Quispe: data curation, validation, visualization, review and editing. **Use of AI:** the authors used Scopus AI (Elsevier) to support the search for seed articles, and Claude (Anthropic) for the search, the analysis code and drafting and editing the text. They reviewed and verified the code, results, references and text, for which they take full responsibility.

References

[1] P. Dollár, C. Wojek, B. Schiele, and P. Perona, "Pedestrian detection: An evaluation of the state of the art," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 4, pp. 743–761, 2012, DOI: 10.1109/TPAMI.2011.155.

[2] Y. P. Loh and C. S. Chan, "Getting to know low-light images with the exclusively dark dataset," *Comput. Vis. Image Underst.*, vol. 178, pp. 30–42, 2019, DOI: 10.1016/j.cviu.2018.10.010.

[3] Z. Zhong, L. Zheng, G. Kang, S. Li, and Y. Yang, "Random erasing data augmentation," in *Proc. AAAI Conf. Artif. Intell.*, vol. 34, no. 7, 2020, pp. 13 001–13 008, DOI: 10.1609/aaai.v34i07.7000.

[4] C. Shang, H. Ai, Z. Zhuang *et al.*, "Improving pedestrian detection in crowds with synthetic occlusion images," in *Proc. IEEE Int. Conf. Multimedia Expo Workshops (ICMEW)*, 2018, pp. 1–4, DOI: 10.1109/ICMEW.2018.8551575.

[5] C. Chi, S. Zhang, J. Xing, Z. Lei, S. Z. Li, and X. Zou, "PedHunter: Occlusion robust pedestrian detector in crowded scenes," in *Proc. AAAI Conf. Artif. Intell.*, vol. 34, no. 7, 2020, pp. 10 639–10 646, DOI: 10.1609/aaai.v34i07.6690.

[6] S. Shao, Z. Zhao, B. Li, T. Xiao, G. Yu, X. Zhang, and J. Sun, "CrowdHuman: A benchmark for detecting human in a crowd," arXiv:1805.00123, 2018.

[7] Instituto Nacional de Estadística e Informática (INEI), "Más de 37 mil efectivos de serenazgo brindaron servicio de seguridad ciudadana," Nota de prensa, 9 abr. 2024, https://www.gob.pe/institucion/inei/noticias/932862, 2024, National Register of Municipalities 2023 (in Spanish).

[8] S. Zhang, L. Wen, X. Bian, Z. Lei, and S. Z. Li, "Occlusion-aware R-CNN: Detecting pedestrians in a crowd," in *Computer Vision – ECCV 2018*. Springer, 2018, pp. 657–674, DOI: 10.1007/978-3-030-01219-9_39.

[9] C. Zhou and J. Yuan, "Bi-box regression for pedestrian detection and occlusion estimation," in *Computer Vision – ECCV 2018*. Springer, 2018, pp. 138–154, DOI: 10.1007/978-3-030-01246-5_9.

[10] X. Song, B. Chen, P. Li, B. Wang, and H. Zhang, "PRNet++: Learning towards generalized occluded pedestrian detection via progressive refinement network," *Neurocomputing*, vol. 482, pp. 98–115, 2022, DOI: 10.1016/j.neucom.2022.01.056.

[11] R. Huan, J. Zhang, C. Xie, R. Liang, and P. Chen, "MLFFCSP: A new anti-occlusion pedestrian detection network with multi-level feature fusion for small targets," *Multimedia Tools and Applications*, vol. 82, no. 19, pp. 29 405–29 430, 2023, DOI: 10.1007/s11042-023-14721-6.

[12] K. Yi, K. Luo, T. Chen, and R. Hu, "An improved YOLOX model and domain transfer strategy for nighttime pedestrian and vehicle detection," *Applied Sciences*, vol. 12, no. 23, 2022, Art. no. 12476. DOI: 10.3390/app122312476.

[13] G. Ghiasi, Y. Cui, A. Srinivas, R. Qian, T.-Y. Lin, E. D. Cubuk, Q. V. Le, and B. Zoph, "Simple copy-paste is a strong data augmentation method for instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2021, pp. 2917–2927, DOI: 10.1109/CVPR46437.2021.00294.

[14] X. Wang, A. Shrivastava, and A. Gupta, "A-Fast-RCNN: Hard positive generation via adversary for object detection," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 3039–3048, DOI: 10.1109/CVPR.2017.324.

[15] C. Michaelis, B. Mitzkus, R. Geirhos, E. Rusak, O. Bringmann, A. S. Ecker, M. Bethge, and W. Brendel, "Benchmarking robustness in object detection: Autonomous driving when winter is coming," arXiv:1907.07484, 2019.

[16] R. Taori, A. Dave, V. Shankar, N. Carlini, B. Recht, and L. Schmidt, "Measuring robustness to natural distribution shifts in image classification," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020.

[17] D. Hendrycks, S. Basart, N. Mu, S. Kadavath, F. Wang, E. Dorundo, R. Desai, T. Zhu, S. Parajuli, M. Guo, D. Song, J. Steinhardt, and J. Gilmer, "The many faces of robustness: A critical analysis of out-of-distribution generalization," in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2021, pp. 8320–8329, DOI: 10.1109/ICCV48922.2021.00823.

[18] E. Mintun, A. Kirillov, and S. Xie, "On interaction between augmentations and corruptions in natural corruption robustness," arXiv:2102.11273, 2021.

[19] K. Kassaw, F. Luzi, L. M. Collins, and J. M. Malof, "Are deep learning models robust to partial object occlusion in visual recognition tasks?" *Pattern Recognition*, vol. 171, p. 112215, 2026, DOI: 10.1016/j.patcog.2025.112215.

[20] R. Pytel, O. S. Kayhan, and J. C. van Gemert, "Tilting at windmills: Data augmentation for deep pose estimation does not help with occlusions," in *Proc. 25th Int. Conf. Pattern Recognit. (ICPR)*, 2021, pp. 10 568–10 575, DOI: 10.1109/ICPR48806.2021.9412475.

[21] S. Cygert and A. Czyżewski, "Toward robust pedestrian detection with data augmentation," *IEEE Access*, vol. 8, pp. 136 674–136 683, 2020, DOI: 10.1109/ACCESS.2020.3011356.

[22] K. Saleh, S. Szénási, and Z. Vámossy, "Occlusion handling in generic object detection: A review," in *Proc. IEEE SAMI*, 2021, pp. 477–484, DOI: 10.1109/SAMI50585.2021.9378657.

[23] A. Kortylewski, J. He, Q. Liu, and A. L. Yuille, "Compositional convolutional neural networks: A deep architecture with innate robustness to partial occlusion," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2020, pp. 8937–8946, DOI: 10.1109/CVPR42600.2020.00896.

[24] A. Wang, Y. Sun, A. Kortylewski, and A. Yuille, "Robust object detection under occlusion with context-aware CompositionalNets," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2020, pp. 12 642–12 651, DOI: 10.1109/CVPR42600.2020.01266.

[25] N. D. Reddy, R. Tamburo, and S. G. Narasimhan, "WALT: Watch and learn 2D amodal representation from time-lapse imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, pp. 9346–9356, DOI: 10.1109/CVPR52688.2022.00914.

[26] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann, "Shortcut learning in deep neural networks," *Nat. Mach. Intell.*, vol. 2, no. 11, pp. 665–673, 2020, DOI: 10.1038/s42256-020-00257-z.

[27] I. Hasan, S. Liao, J. Li, S. U. Akram, and L. Shao, "Generalizable pedestrian detection: The elephant in the room," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2021, pp. 11 323–11 332, DOI: 10.1109/CVPR46437.2021.01117.

[28] V. S. Sukesh Babu and R. Raman, "Robust pedestrian detection via curated training on created dataset," in *Proc. IEEE 21st India Council Int. Conf. (INDICON)*, 2024, pp. 1–6, DOI: 10.1109/INDICON63790.2024.10958520.

[29] Z. Cui, G.-J. Qi, L. Gu, S. You, Z. Zhang, and T. Harada, "Multitask AET with orthogonal tangent regularity for dark object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2021, pp. 2553–2562, DOI: 10.1109/ICCV48922.2021.00255.

[30] L. Neumann, M. Karg, S. Zhang, C. Scharfenberger, E. Piegert, S. Mistr, O. Prokofyeva, R. Thiel, A. Vedaldi, A. Zisserman, and B. Schiele, "NightOwls: A pedestrians at night dataset," in *Computer Vision – ACCV 2018*, vol. 11361, 2019, pp. 691–705, DOI: 10.1007/978-3-030-20887-5_43.

[31] S. Zhang, R. Benenson, and B. Schiele, "CityPersons: A diverse dataset for pedestrian detection," in *Proc. IEEE Conf. Com-*

put. Vis. Pattern Recognit. (CVPR), 2017, pp. 3213–3221, DOI: 10.1109/CVPR.2017.474.

[32] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, pp. 10 674–10 685, DOI: 10.1109/CVPR52688.2022.01042.

[33] L. Yang, B. Kang, Z. Huang, Z. Zhao, X. Xu, J. Feng, and H. Zhao, "Depth anything V2," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 37, 2024.

[34] Ultralytics, "Ultralytics YOLO11 documentation," https://docs.ultralytics.com/models/yolo11, 2024, Version 8.4.143. Accessed: Jun. 2026.

[35] ——, "Ultralytics YOLO26," https://docs.ultralytics.com/models/yolo26/, 2026, Accessed: Oct. 20, 2026.

[36] Y. Zhao, W. Lv, S. Xu, J. Wei, G. Wang, Q. Dang, Y. Liu, and J. Chen, "DETRs beat YOLOs on real-time object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 16 965–16 974, DOI: 10.1109/CVPR52733.2024.01605.

[37] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, 2017, DOI: 10.1109/TPAMI.2016.2577031.

[38] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, 2017, pp. 2999–3007, DOI: 10.1109/ICCV.2017.324.

[39] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick, "Microsoft COCO: common objects in context," in *Computer Vision – ECCV 2014*, vol. 8693, 2014, pp. 740–755, DOI: 10.1007/978-3-319-10602-1_48.

[40] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. New York: Chapman & Hall/CRC, 1993.

[41] D. Lakens, "Equivalence tests: A practical primer for t tests, correlations, and meta-analyses," *Soc. Psychol. Personal. Sci.*, vol. 8, no. 4, pp. 355–362, 2017, DOI: 10.1177/1948550617697177.

[42] M. Oquab, T. Darcet, T. Moutakanni, H. Vo, M. Szafraniec, V. Khalidov *et al.*, "DINOv2: Learning robust visual features without supervision," *Trans. Mach. Learn. Res.*, 2024.